MIL-V-00500660(MU)
14 July 1967

USED IN LIEU OF
MIL-V-50066B
30 September 1964

MILITARY SPECIFICATION

VOICEMITTER-OUTLET VALVE ASSEMBLY, C29R1

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-V-50066B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of protective mask voicemitter-outlet valve assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

US ARMY MUNITIONS COMMAND

DL5-81-3 - Voicemitter-Outlet Valve, C29R1.

(All drawings referenced in this specification are listed on above Edgewood Arsenal drawing.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-81-3 or its subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the voicemitter-outlet valve shall conform to the specifications listed on Drawing DL5-81-3 and its subsidiary drawings.

3.2 Manufacture and assembly. The voicemitter-outlet valve shall be manufactured and assembled to conform to Drawing D5-81-3 and subsidiary drawings listed thereon.

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3.3 Diaphragm assembly.

3.3.1 DOP smoke penetration. The diaphragm assembly shall show a DOP smoke penetration of no more than 0.001 percent when tested as specified in 4.3.4.1.

3.3.2 Diaphragm tension. Tension on the diaphragm shall be such that the center of the diaphragm shall be deflected no less than 0.025 inch nor more than 0.040 inch, without seating on the baffle, when tested as specified in 4.3.4.2.

3.3.3 GB penetration of diaphragm material. The diaphragm material shall resist the penetration of vapor GB for no less than two hours when tested as specified in 4.3.4.3. The initial 15 linear feet of each roll of diaphragm material used for preproduction or regular production shall be submitted to the Government for GB testing.

3.4 Resistance to airflow. The assembled outlet valve shall have a resistance not exceeding a pressure differential of 22 millimeters of water at an airflow rate of 85 liters per minute when tested as specified in 4.3.4.4.

3.5 Outlet valve leakage. Leakage through the assembled outlet valve shall not exceed 15 milliliters of air per minute, at a pressure differential of one inch of water, when tested as specified in 4.3.4.5.

3.6 DOP smoke penetration. DOP smoke penetration of the voicemitter-outlet valve assembly with the opening in the valve seat sealed shall not exceed 0.001 percent when tested as specified in 4.3.4.1.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of voicemitter-outlet valve assemblies shall be produced in accordance with this specification for examination and tests (see 4.2). Any change in materials or operating conditions shall require a new preproduction sample.

3.8 Workmanship. The voicemitter-outlet valve assembly shall be free from foreign matter, and damage such as distortion, dents, roughened valve seat, and torn disc that would impair its function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the testing in 4.3.4.3. Sample diaphragm material shall be forwarded to the laboratory designated by the contracting officer (see 6.2).

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 10 voicemitter-outlet valve assemblies shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.2.2 Inspection.

4.2.2.1 For examination. The sample voicemitter-outlet valve assemblies shall be inspected for all requirements of the drawings and this specification.

4.2.2.2 For tests. The sample voicemitter-outlet valve assemblies shall be tested in accordance with paragraph 4.3.4.

4.2.3 Acceptance/rejection criteria. The preproduction sample shall meet the examinations and tests specified by 4.2.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

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4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the voicemitter-outlet valve assemblies produced by one manufacturer, at one plant, from the same materials, and under the same manufacturing conditions. No more than one lot of outlet valve discs shall be represented in any one lot of voicemitter-outlet valve assemblies.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection.

4.3.3.1 For examination and nondestructive tests. Sample voicemitter-outlet valve assemblies shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.3.3.2 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.3.3.3 Classification of defects.

(a) Diaphragm assembly (Dwg C5-81-13).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration	4.3.4.1
Major:	AQL 0.25 percent defective	
101	Diaphragm tension	4.3.4.2
	AQL 1.0 percent defective	
102	Component missing or incorrectly assembled	
103	Outside diameter (2.364) incorrect	
104	Outside periphery thickness (0.130) incorrect	
105	Workmanship (see 3.8)	

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4.3.3.3 Classification of defects. (Continued.)

(b) Voicemitter-outlet valve, C29P1, assembly (Dwg D5-81-3).

Categories	Defects	Acceptance standards
Critical:		
1	Outlet valve leakage	4.3.4.5
2	DOP smoke penetration	4.3.4.1
Major:	AQL 0.25 percent defective	
101	Outlet valve resistance to airflow	4.3.4.4
	AQL 1.0 percent defective	
102	Component missing, incorrect, or incorrectly assembled	
103	Sealing ring loose	
104	Workmanship (see 3.8)	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing	

4.3.4 Tests. Tests shall be conducted as follows:

4.3.4.1 DOP smoke penetration. Determine the leakage of the voicemitter-outlet valve assembly or the diaphragm assembly with Q127 Penetrometer.

4.3.4.2 Diaphragm tension. Determine the diaphragm tension with Q41 Tension Tester. Apply 114 ± 1 grams of weight on the center of the diaphragm by means of a rod which passes freely through the center hole of the cupped baffle. The total weight of the rod, with attached linkage, shall be 19 ± 1 gram. Measure only the deflection of the diaphragm caused by the addition of the 114 gram weight. Use an additional 1 to 5 gram weight to determine that the diaphragm has not seated or bottomed on the baffle.

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4.3.4.3 GB penetration. Determine the resistance of the diaphragm material to GB penetration by use of the Q171 Static Diffusion Apparatus.

4.3.4.4 Resistance to airflow. Test the outlet valve assembly of the voicemitter for resistance to airflow with Q106 Inlet and Outlet Valve Resistance Indicator.

4.3.4.5 Outlet valve leakage. Test voicemitter-outlet valve assembly for leakage with M4A1 Outlet Valve Leakage Indicator.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The voicemitter-outlet valve assemblies shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Voicemitter-outlet valve assemblies, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any Special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is prepared for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is indentified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. The item covered by this specification is a combination voicemitter and outlet valve, intended for use with the ABC-M17 protective mask.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Component specifications. Following are listed the component specifications for the C29R1 voicemitter-outlet valve:

Discs, Outlet Valve - MIL-D-10132.
Voicemitter-Outlet Valve, C29R1, Components For - MIL-V-50067.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 21, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:

Army - MU

Preparing activity:

Army - MU (EA)

International interest (see 6.4)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

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